Mid Semester Examination 2025

Name of the Program: B.Tech. Civil Engineering — *Semester:* V

Name of the Course: Hydrology and Irrigation Engineering — *Course Code:* TCE-504

Time: $1\frac{1}{2}$Hour — Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

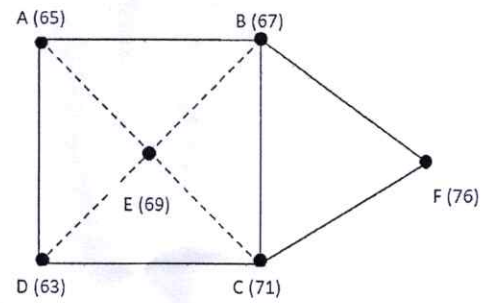
Q1	(10 marks)	CO-1
(a)	Draw a neat sketch a natural siphon type of rain gauge and explain its working.	
	OR	
(b)	Find the mean precipitation for the area sketched below by Theisson method. The area is composed of a square plus an equilateral triangular plot of side 10 kms. Rainfall readings in mm at the various stations are given. Check the same by arithmetic average method. 	
Q2	(10 marks)	CO-1
(a)	Describe the methods to reduce evaporation loss from reservoirs.	
	OR	
(b)	Station “X” failed to report the rainfall recorded during a storm. With respect to east- west and north-south axes set up at station “X”, the coordinates of 4-surrounding gauges, which are nearest to station “X” in respective quadrants are (10,15), (-10,5), (-5,-10) and (5,-15) km respectively. Determine the missing rainfall at station “X”, if the storm rainfalls at the four surrounding gauges are 75, 90, 70 and 55 mm respectively.	
Q3	(10 marks)	CO-2
(a)	What is runoff? Discuss various factors which affect runoff from a catchment.	
	OR	
(b)	The ordinates of a 2-h UH for a basin are given. Derive the flood hydrograph due to a 2-h storm producing ER of 4 cm. Assume constant base flow of 1 cumec.	
Q4	(10 marks)	CO-2
(a)	Draw a typical single peaked flood hydrograph and described the various parts of it	
	OR	

Table for Q3 (b):

Time (hr)	0	2	4	6	8	10	12	14	16	18
2-hr UHGO	0	1.5	4.5	8.6	12	9.4	4.6	2.3	0.8	0

(b) The ordinates of a 4-hr UH for a particular basin are given below, Derive the ordinates of i) S-curve hydrograph, ii) Area of basin

Time(hr)	0	2	4	6	8	12	14	16	18	20	22	24
4-hr UHGO	0	25	100	160	190	170	110	70	30	8	3	0

Q5 (10 marks)

CO-1 & CO-2

(a) The average rainfall over a basin of area of 50 hectares during a storm was follows

Time (hr)	0	1	2	3	4	5	6	7
Rainfall(mm)	0	6	11	34	28	12	6	0

If the volume of runoff from this storm was measured as 25×10^3 m^3. Determine the φ-index for the storm

OR

(b) Given below are the observed flows from a storm of 3- hr duration on a stream with a drainage area of 40 km^2.
Derive the 3 hr unit hydrograph for the catchments and plot the same.

Time (hr)	2	5	8	11	14	17	20	23	26	29	32	35	38	41	44	47
Q, m3/s	50	47	75	120	225	290	270	145	110	90	80	70	60	55	51	50
BFO m3/s	50	47	47	47	47	48	48	48	48	49	49	50	50	55	51	50